



Universidad Tecnológica de Bolívar

Assignment 4 (04/11/2026)

Undetermined coefficients vs. (forward) Euler method for the solution of 2nd order nonhomogeneous linear ODEs with constant coefficients

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The purpose of this problems is to analyze the effectiveness of Forward Euler method in approximating the solution of an initial value problem involving a 2nd order nonhomogeneous linear ODE with constant coefficients.

Problem 1

This is an example problem taken from [Zill D. \(2014\)](#):

Consider the differential equation

$$y'' + 4y' + 4y = (3 + x)e^{-2x}.$$

- (0.5 pts.) Find a general solution for the equation by undetermined coefficients.
- (0.5 pts.) Given the initial conditions $y(0) = 2$ and $y'(0) = 5$, find the solution for the IVP.

Problem 2

- (a) Consider the interval $[0, 5]$, a discretization of the interval with constant step size $h = 0.5$ (i.e., $x_0 = 0.0$, $x_1 = 0.5$, $x_2 = 1.0$, $\dots$, $x_{10} = 5.0$) and the IVP in **Problem 1**.
- (i) (0.5 pts.) Using central finite difference about x_1 and the initial conditions, find the value $y_1 \approx y(x_1)$. Using finite difference for the 2nd order derivative, build an iterative scheme on the index i to determine the value $y_i \approx y(x_i)$ for $i = 2, 3, \dots, 10$.
 - (ii) (1.0 pts.) Determine the local and global truncation errors expected for the pre-

vious approach according to the theory provided for the error analysis.

- (iii) (1.0 pts.) Implement your algorithm using a Python-supported platform (Anaconda, Google Colab or a local compiler). Then, run your code for different values of $h \leq 0.5$ (suggested, halve the value of h after every run). Plot h vs. E_l and h vs. E_g , where E_l and E_g are the local and global truncation errors obtained, respectively (use a log-log plot and observe the slope of the resulting line). Compare with the local/global error found in the analysis of **Problem 2** (iii).

Problem 3

This is an example problem taken from [Zill \(2024\)](#):

- (a) (0.5 pts.) A projectile shot from a gun has weight $w = mg$ and velocity v tangent to its path of motion. ignoring air resistance and all other forces acting on the projectile except its weight, determine a system of differential equations that describes its path of motion. See Figure 1. Solve the system (Hint: use Newton's second law of motion in x and y directions).
- (b) (1.0 pts.) Write a function `my_projectile-motion_solver(t_span,s0,m,g)` that solves the projectile motion problem using `solve_ivp` (from the SciPy library in Python). The function returns `[T,X,Y]`. The input argument `t_span` should be a list of the form `[t0, tf]`, where t_0 is the initial time, and t_f is the final time of consideration. The input argument `s0` should be a 2D array of the form `[x0, y0]`, where (x_0, y_0) represents an initial position. Finally, the input arguments `m` and `g` are scalar parameters denoting the mass of the projectile and the gravity force, respectively. The output argument `T` should be an array of times given as the output argument of `solve-ivp`. The output arguments `X` and `Y` should be the numerically integrated solution of the problem in part (a).

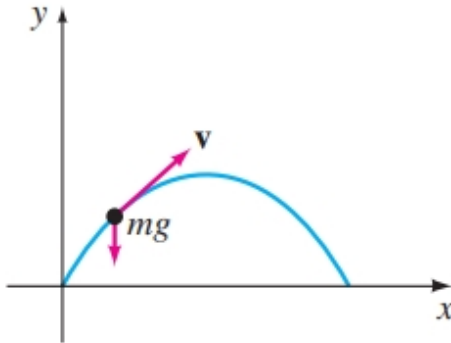


Figure 1: Path of the projectile in Problem 3.

References

Zill, D. (2024). *A First Course in Differential Equations with Modeling Applications, Metric Version*. Cengage.

Zill D., W. W. (2014). *Advanced Engineering Mathematics*. Jones & Bartlett Learning.